

Data Analytics Capstone

Machine-Learning Forecasting of Solar Irradiance and Reliability-Constrained Off-Grid Solar Sizing Across Ghana's Climatic Zones

Interim Report

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GitHub Repository

All code, raw and processed data, notebooks, figures, and results tables referenced in this report are publicly available at: <https://github.com/kudumens/capstone-solar-ghana> (Adjei-Mensah, 2026). The repository README documents one-command reproduction of the full pipeline (see Appendix A).

Introduction

Background and Context

Ghana's electricity supply has traditionally relied on hydropower from the Volta basin and thermal generation fueled by imports, leaving the grid vulnerable to fluctuations in rainfall and fuel prices. The Government of Ghana's Renewable Energy Master Plan, adopted in 2019, set a target to increase renewable generation capacity from a 2015 baseline of 42.5 MW to approximately 1,364 MW by 2030, including about 500 MW of utility-scale solar and 200 MW of distributed solar photovoltaics (PV; Energy Commission of Ghana, 2019). Ghana's tropical climate provides abundant solar energy, with daily global horizontal irradiance (GHI) of approximately 4–6 kWh/m²/day and a distinct north–south gradient that favors the savannah regions (Asumadu-Sarkodie & Owusu, 2016).

Two groups would benefit substantially from accurate resource forecasting. First, grid operators and developers, including the Ghana Grid Company (GRIDCo) and the Electricity Company of Ghana (ECG), need short- and medium-term irradiance forecasts for generation planning, reserve capacity, and plant siting. Second, households, often overlooked in research, are increasingly adopting off-grid solar in response to unreliable grid power and falling PV and battery costs, yet they lack an evidence-based method for sizing installations. Installers commonly apply rules of thumb, such as a fixed number of days of autonomy, that ignore

regional variation; for example, the elevated risk of consecutive low-sun days in coastal Accra during the rainy season compared with Navrongo in the Sudan savannah during the Harmattan.

Machine-learning (ML) methods now dominate the international solar-forecasting literature (Ahmed et al., 2020; Lauret et al., 2015; Voyant et al., 2017), and deep-learning models such as long short-term memory (LSTM) networks have achieved 18–34% error reductions over baselines in tropical settings (Qing & Niu, 2018). For Ghana specifically, Quansah et al. (2022) validated NASA Prediction of Worldwide Energy Resources (POWER) solar-radiation estimates against ground-station-based estimates at 22 Ghanaian synoptic stations ($r = .60\text{--}.94$), supporting NASA POWER as a long-term solar data source for regional comparison. Ghana-specific ML research on solar forecasting remains limited to plant-level studies such as Asiedu et al. (2024), and the literature review conducted for this project identified no published work linking forecasting-grade irradiance modeling to household system sizing.

Problem Statement

Solar resource uncertainty in Ghana is currently managed with simple averages and heuristics: Grid planners lack systematic evidence about which forecasting models are most effective across climatic zones and seasons, and off-grid households size panels and batteries with fixed rules that can waste capital through oversizing or cause blackouts through undersizing. This project addresses two gaps. First, no published study systematically compares classical and ML irradiance forecasts across Ghana’s coastal, forest, and savannah zones using a common long-term dataset and formal forecast-comparison tests. Second, no publicly documented, reproducible tool converts Ghana’s historical irradiance record into reliability-constrained, zone-specific off-grid sizing guidance: the number of panels and the battery capacity needed to meet a specified consumption level at a given reliability level year-round.

Purpose of the Study

The study pursues two related aims: prediction and optimization. First, it predicts daily GHI by developing and statistically evaluating classical models (persistence, day-of-year climatology, and SARIMA-class dynamic harmonic regression) and ML models (random forest, XGBoost, and LSTM) at five locations spanning Ghana's three climatic zones, using 20 years (2005–2024) of NASA POWER daily data, and it explains the meteorological drivers of irradiance using correlation analysis, permutation importance, and Shapley additive explanations (SHAP). Second, it optimizes by transforming the same 20-year record into a reliability-constrained off-grid sizing model that identifies the smallest feasible PV array and battery configurations that satisfy predefined reliability constraints through historical energy-balance simulation.

Interim Project Status (Progress Snapshot)

As of August 3, 2026:

Completed:

- Data collection: 20 years (2005–2024) of daily NASA POWER data downloaded for all five sites ($7,305 \text{ days} \times 5 \text{ sites} = 36,525 \text{ site-days}$), with SHA-256 checksums logged.
- Data validation: row counts, calendar continuity, duplicate checks, missing-value screening, and physical-range checks: all sites pass with zero missing values (notebook 02).
- Data cleaning and feature engineering: sentinel handling, continuous daily indexing, GHI lags 1–7, 7- and 30-day rolling means, cyclical day-of-year encoding, and season labels (src/clean_data.py).
- Exploratory data analysis, including univariate and bivariate distribution analysis: GHI histograms, driver statistics, a pair plot of all variable combinations, seasonal cycles, distributions by site and season, driver correlations, interannual variability, and consecutive low-sun-day analysis (notebook 03).

- Classical baselines: persistence, day-of-year climatology, and dynamic harmonic regression fitted per site and evaluated on the 2021–2022 validation window at 1- and 7-day horizons (notebook 04).
- Preliminary ML models: random forest and XGBoost at the 1-day horizon with time-series cross-validated tuning, evaluated against the classical baselines (notebook 05, partial).

In progress:

- LSTM model (30-day input windows); extension of ML models to the 7-day horizon; broader hyperparameter search.

Pending:

- Final test-set (2023–2024) evaluation with Diebold–Mariano significance tests (RQ1); SHAP feature-importance analysis (RQ2); seasonal and zonal error stratification (RQ3); off-grid PV–battery sizing simulation (RQ4); recommendations.

Scope and Objectives

The project is bounded to five study sites representing Ghana’s climatic zones; Accra (coastal savannah), Takoradi (coastal), Kumasi (moist forest), Tamale (Guinea savannah), and Navrongo (Sudan savannah), at daily resolution over 2005–2024, with forecast horizons of 1 day and 7 days ahead under a direct forecasting strategy in which all predictors are restricted to information available at the forecast origin. Sub-daily forecasting, plant-level power telemetry, and primary data collection are out of scope. The five sites were selected because together they span Ghana’s three recognized climatic zones and the full north–south irradiance gradient; because NASA POWER provides spatially smoothed gridded data at roughly 0.5° resolution, additional sites within the same zone would contribute little independent climatic information, so the marginal analytical value of further sites is low. The forecasting comparison (Research Questions 1–3) is the primary contribution and the sizing application (Research Question 4) the

secondary contribution. These bounds translate into four measurable objectives, each paired with a research question below:

Objective 1 (O1): Train and evaluate six forecasting models per site and demonstrate statistically significant skill ($\text{Skill} > 0$) over the classical baselines on the untouched 2023–2024 test window at both the 1-day and 7-day horizons.

Objective 2 (O2): Rank the five meteorological drivers of daily GHI using correlation tests, permutation importance, and SHAP values, reporting Holm-corrected significance for each driver.

Objective 3 (O3): Quantify wet–dry-season and cross-zone differences in forecast accuracy using formal two-sample and multigroup tests at $\alpha = .05$.

Objective 4 (O4): Produce zone-specific PV–battery sizing tables that meet Loss-of-Load Probability (LOLP) targets of 1% and 5% for the 100, 200, and 300 kWh/month consumption tiers.

The study pursues four research questions:

Research Question 1 (RQ1): How accurately can machine-learning models (random forest, XGBoost, LSTM) forecast daily global horizontal irradiance at representative sites across Ghana’s coastal, forest, and savannah climatic zones, and do they significantly outperform classical approaches (persistence, day-of-year climatology, and dynamic harmonic regression) at the 1-day and 7-day forecast horizons?

Research Question 2 (RQ2): Which meteorological variables (temperature, relative humidity, cloud amount, precipitation, wind speed) are most strongly associated with daily solar-irradiance variation in Ghana, and which contribute most to predictive performance?

Research Question 3 (RQ3): How does forecasting accuracy differ between the wet and dry seasons (including the Harmattan) and across the north–south climatic gradient?

Research Question 4 (RQ4): What is the minimum feasible PV-array and battery configuration that satisfies predefined reliability constraints (loss-of-load probability [LOLP] $\leq 1\%$ and $\leq 5\%$) for representative household electricity demands (100, 200, and 300 kWh/month) across Ghana’s climatic zones year-round?

The two LOLP targets correspond to 99% and 95% supply reliability, the range conventionally used in standalone PV sizing studies (Khatib et al., 2013). Reporting both brackets the cost-reliability trade-off: 1% (about 3 to 4 unserved days per year) represents a near-firm household supply, while 5% represents a budget-constrained design, and the gap between the two configurations quantifies the price of the last few percentage points of reliability.

The 100, 200, and 300 kWh/month tiers have been chosen to span low-consumption rural to high-consumption urban households around the national average of roughly 135 kWh per residential customer per month (Energy Commission of Ghana, 2025).

Research Hypotheses

Each research question is paired with a directional hypothesis, tested at $\alpha = .05$;

Hypothesis 1 (H1): Machine-learning models significantly outperform the classical benchmarks for 1-day-ahead GHI forecasts in each climatic zone ($\text{Skill} > 0$) and reduce RMSE by at least 15% relative to the day-of-year climatology baseline, with R^2 reported descriptively rather than tested against a fixed threshold. The 15% threshold is set conservatively based on error reductions in deep-learning solar forecasts (18–34%; Qing & Niu, 2018), applied against a stronger reference than baseline methods. It also acts as a materiality criterion: with about 3,650

test days, smaller differences ($d = 0.10$) would be statistically detectable, so the threshold filters out improvements too minor to matter for planning.

Hypothesis 2 (H2): Cloud amount and relative humidity show significantly stronger associations with daily GHI than temperature, precipitation, or wind speed.

Hypothesis 3 (H3): Forecast RMSE differs significantly between wet and dry seasons, with the largest degradation in the southern zones during June–August.

Hypothesis 4 (H4): Required battery capacity for a fixed reliability target differs across climatic zones by at least 25%, implying that uniform national sizing rules would materially mis-size systems. The 25% threshold is primarily a materiality criterion, not based on detectability. Block-bootstrap intervals spanning 20 years can identify smaller zone differences, so the cutoff point is where differences impact purchasing decisions. For the 200 kWh/month tier (around 16.7 kWh of usable storage), 25% equals about 4 kWh, exceeding a single battery module. Consequently, a national rule with this margin could lead to stranded capital in northern regions or reliability issues in the south. Studies show that results greatly vary by location (Khatib et al., 2013), and analysis indicates this threshold is cautious: low-sun day runs differ by a factor of two between north (3 days) and south (6–7 days), with autonomy scaling accordingly.

Literature Survey

Literature Review Approach

Sources were identified through Google Scholar, Scopus, ScienceDirect, and IEEE Xplore using combinations of the keywords “Ghana,” “solar irradiance forecasting,” “machine learning,” “NASA POWER,” “off-grid PV sizing,” and “loss-of-load probability,” for publications through July 2026. Inclusion favored (a) peer-reviewed studies of solar forecasting with ML or classical time-series methods, with tropical settings weighted most heavily; (b)

methodological primary sources for each model and statistical test used; (c) validation studies of the NASA POWER data source in Ghana; (d) Ghanaian energy-policy and consumption statistics defining stakeholders and the demand benchmark; and (e) reviews of off-grid PV sizing. The 19 sources fall into five bodies of literature aligned with the research questions: forecasting methods (RQ1), forecast-evaluation statistics (RQ1/RQ3), data-source validation, the Ghanaian energy context, and PV system sizing (RQ4).

Summary of Key Literature

Each source below is paraphrased with its purpose, method, findings, and relevance to this project. Table 1 then maps the full set to the research questions.

Ahmed et al. (2020) systematically review the landscape of PV solar-power forecasting: statistical, machine-learning, deep-learning, and hybrid techniques, cataloging evaluation metrics and preprocessing strategies. They conclude that no single method dominates all horizons: tree ensembles and neural networks excel at short horizons while classical methods remain competitive for longer aggregation. This framework informs the project's model selection and its decision to evaluate both 1-day and 7-day horizons.

Voyant et al. (2017) review machine-learning methods for solar-radiation forecasting, comparing regressions, neural networks, support-vector machines, and tree ensembles, and emphasize rigorous out-of-sample testing against naive baselines. The review anchors the study's requirement that every ML model be benchmarked against classical references before any claim of skill.

Lauret et al. (2015) benchmark neural networks, Gaussian processes, and support-vector regression against classical autoregressive baselines for hourly solar forecasting on tropical islands, finding that ML delivers its largest gains under unstable, cloudy skies while simple

baselines remain competitive under clear skies. This insight, that the value of ML depends on sky-condition regime, maps directly onto RQ3 and onto this report’s preliminary finding that ML gains are largest at Ghana’s cloudier southern sites.

Qing and Niu (2018) apply LSTM networks to day-ahead irradiance prediction in a tropical setting (Santiago de Cuba), reporting 18–34% RMSE improvements over multilayer-perceptron and persistence baselines. The paper is the methodological precedent for the project’s LSTM component; notably, its effect sizes are computed against persistence-type baselines, which contextualizes the more modest margins observed here against stronger classical references (see Preliminary Results).

Asiedu et al. (2024) compare single and hybrid ML models (ANN, random forest, XGBoost) for forecasting production of a 180-kWp grid-connected PV plant at Tema, Ghana, finding ANN best day-ahead and ensemble/hybrid models best at longer horizons. As the closest Ghanaian precedent, it defines the research gap: Ghanaian ML solar forecasting exists but is confined to a single coastal industrial site, with no multi-zone comparison, no seasonal stratification, and no household application, precisely the dimensions this capstone adds.

Quansah et al. (2022) assess NASA POWER solar-radiation estimates against ground-based estimates derived from sunshine-duration observations at 22 Ghanaian synoptic stations, reporting correlations of $r = .60\text{--}.94$ with strongest agreement in the northern savannah. The study justifies NASA POWER as this project’s data source while motivating the explicit site-level bias examination performed during EDA.

Asumadu-Sarkodie and Owusu (2016) document Ghana’s solar-resource endowment (roughly 4–6 kWh/m²/day) and its north–south gradient, arguing that solar PV is a viable complement to hydro-dominated generation. The paper establishes the domain motivation and

justifies the choice of study sites spanning the coastal, forest, and savannah zones: a gradient this project’s EDA reproduces quantitatively.

Breiman (2001) introduces random forests and proves that generalization error converges as trees are added, making the method resistant to overfitting. Random forests handle nonlinear interactions among the engineered predictors without distributional assumptions (RQ1), and the paper’s variable-importance machinery is the ancestor of the importance analyses used for RQ2. Its known inability to extrapolate beyond training targets motivates the seasonal error stratification planned for RQ3.

Chen and Guestrin (2016) present XGBoost, the regularized gradient-boosted tree system that is a de facto standard for structured-data prediction. Engineered tabular features: lags, rolling means, cyclical calendar encodings, meteorological drivers, are exactly the input type where boosted trees are most competitive, and the paper’s regularization discussion informs the tuning grid used in notebook 05.

Hochreiter and Schmidhuber (1997) introduce the LSTM architecture whose gated memory cell solves the vanishing-gradient problem, enabling recurrent networks to learn long temporal dependencies such as multi-day weather episodes. It is the primary source for the project’s pending deep-learning forecaster; the study does not presume LSTM superiority given roughly 5,800 training days per site, hence the planned pooled multi-site variant and repeated seeded runs.

Hyndman and Athanasopoulos (2021) provide the statistical foundations for the classical baselines and evaluation design: seasonal-naive methods, SARIMA modeling, dynamic harmonic regression for long seasonal periods, temporal train/test splits, and time-series cross-

validation. The text directly shapes the temporal-split design and the implementation of the dynamic harmonic regression baseline reported below.

Diebold and Mariano (1995), with the Harvey et al. (1997) small-sample correction, supply the formal machinery for comparing forecast accuracy on the same test days while accounting for the autocorrelation of forecast errors. These tests will guide RQ1’s model-versus-baseline comparisons on the untouched 2023–2024 test window, complemented by moving-block bootstrap intervals (Künsch, 1989).

Lundberg and Lee (2017) introduce SHAP, which unifies feature-attribution methods as approximations of Shapley values and provides efficient estimators for tree ensembles. SHAP will rank predictor contributions for RQ2, interpreted as model-specific predictive contributions rather than causal effects.

Khatib et al. (2013) review photovoltaic system-sizing techniques and frame reliability through the loss-of-load-probability concept, showing that numerical simulation over long historical irradiance records is the most accurate sizing approach and that outcomes are highly location-dependent. This is the methodological foundation for RQ4’s planned 20-year energy-balance backtest, and the source of the performance-ratio and depth-of-discharge constants adopted in the design.

Energy Commission of Ghana (2019, 2025) provides the policy roadmap (renewable-capacity targets through 2030) and the national consumption statistics (residential consumption of roughly 8,569 GWh across about 5.28 million residential customers in 2024, approximately 135 kWh per customer per month) that define the stakeholder context and ground the 100/200/300 kWh/month demand tiers used in RQ4.

Table 1

Literature Relevance Matrix

Author (Year)	Domain/Context	Dataset/Setting	Method(s)	Key Findings	Supports RQ(s)
Ahmed et al. (2020)	Solar forecasting review	Global literature	Systematic review	No single method wins all horizons	RQ1, RQ3
Voyant et al. (2017)	ML solar forecasting review	Global literature	Review	Benchmark ML vs. naive baselines	RQ1
Lauret et al. (2015)	Tropical islands	Hourly GHI, 3 islands	ANN, GP, SVR vs. AR	ML gains largest in cloudy regimes	RQ1, RQ3
Qing & Niu (2018)	Tropical day-ahead	Hourly, Santiago de Cuba	LSTM	18–34% RMSE gain over persistence/MLP	RQ1
Asiedu et al. (2024)	Ghana PV plant	180 kWp plant, Tema	ANN, RF, XGBoost, hybrids	Ghanaian ML forecasting exists; single site only	Gap, RQ1
Quansah et al. (2022)	Data validation, Ghana	22 synoptic stations	NASA POWER vs. ground	$r = .60-.94$; best in north	Data validity
Asumadu-Sarkodie & Owusu (2016)	Ghana solar potential	National resource data	Techno-economic assessment	4–6 kWh/m ² /day; N–S gradient	Motivation, RQ3
Breiman (2001)	ML methodology	—	Random forests	Ensemble resists overfitting	RQ1, RQ2
Chen & Guestrin (2016)	ML methodology	—	Gradient boosting	Regularized boosting for tabular data	RQ1, RQ2
Hochreiter & Schmidhuber (1997)	Deep learning	—	LSTM	Gated memory learns long dependencies	RQ1
Hyndman & Athanasopoulos (2021)	Forecasting text	—	Classical TS methods	DHR for long seasonality; temporal CV	RQ1, design
Diebold & Mariano (1995); Harvey et al. (1997)	Forecast comparison	—	DM test + HLN correction	Valid tests under autocorrelation	RQ1
Künsch (1989)	Inference	—	Moving-block bootstrap	Valid intervals for dependent data	RQ1, RQ4
Lundberg & Lee (2017)	Interpretability	—	SHAP	Unified additive attributions	RQ2
Khatib et al. (2013)	Off-grid PV sizing	Review	LOLP simulation methods	Historical simulation most accurate; location-dependent	RQ4
Energy Commission of Ghana (2019, 2025)	Ghana policy/statistics	National	Policy & statistics	Targets; ~135 kWh/customer/month	Motivation, RQ4

Thematic Synthesis

Across the forecasting literature, three themes converge on this project’s design. First, model choice is horizon- and regime-dependent: reviews (Ahmed et al., 2020; Voyant et al., 2017) and tropical benchmarks (Lauret et al., 2015) agree that ML gains concentrate in short

horizons and unstable skies, which motivates this study’s dual 1-/7-day design and its seasonal stratification (RQ3). Second, honest evaluation requires strong baselines and dependence-aware statistics: the persistence and climatology references, the dynamic harmonic regression of Hyndman and Athanasopoulos (2021), and the Diebold–Mariano/HLN/bootstrap machinery together guard against the inflated skill claims that arise when ML is compared only with weak naive baselines, a caution that this report’s preliminary results have already vindicated.

Third, the Ghanaian literature is thin precisely where this project operates. Asiedu et al. (2024) demonstrate feasibility at a single coastal plant; Quansah et al. (2022) validate the data source but do not forecast; and the policy sources quantify demand without linking it to resource variability. No identified study compares models across Ghana’s climatic zones on a common long-term record, and none converts forecasting-grade resource analysis into reliability-constrained household sizing. The sizing literature (Khatib et al., 2013) supplies the LOLP simulation method but predates modern ML and contains no Ghana-specific, zone-differentiated guidance. The capstone therefore occupies a defensible gap at the intersection of these bodies of literature.

Data Description

Data Source(s) and Access

The primary data source is the NASA POWER archive (NASA Langley Research Center, 2024), which provides daily satellite-derived solar and meteorological parameters on a global grid, free of charge and without registration, via a documented REST API. Data quality for Ghana is supported by Quansah et al. (2022). Archive and documentation: <https://power.larc.nasa.gov/>. Example API call (Accra, all six parameters, 2005–2024): https://power.larc.nasa.gov/api/temporal/daily/point?parameters=ALLSKY_SFC_SW_DWN,T2

M,RH2M,CLOUD_AMT,PRECTOTCORR,WS2M&start=20050101&end=20241231&latitude=5.60&longitude=-0.19&community=RE&format=CSV

Dataset Overview

- Number of records (rows): 7,305 daily observations per site $\times$ 5 sites = 36,525 site-days (raw); the processed modeling dataset (all_sites_daily.csv) holds the same 36,525 rows with engineered features.
- Number of variables (columns): 6 raw meteorological parameters; 21 columns after feature engineering (site, zone, season, cyclical calendar encodings, GHI lags 1–7, 7-/30-day rolling means).
- Time period: January 1, 2005 – December 31, 2024 (20 years, daily resolution).
- Unit of analysis: the site-day.
- Target variable: ALLSKY_SFC_SW_DWN — daily all-sky global horizontal irradiance (GHI), kWh/m²/day.

The five study sites span Ghana’s three recognized climatic zones and the full north–south irradiance gradient: Accra (coastal savannah; 5.60 N, 0.19 W), Takoradi (coastal; 4.90 N, 1.76 W), Kumasi (moist semi-deciduous forest; 6.69 N, 1.62 W), Tamale (Guinea savannah; 9.40 N, 0.84 W), and Navrongo (Sudan savannah; 10.89 N, 1.09 W).

The adequacy of this volume for each research question is verified formally in the Analysis section (Minimum Sample Size Computation per Research Question, Table 6); the binding requirement of 1,537 daily observations per site is exceeded more than fourfold.

Table 2

Data Dictionary

Variable	Definition	Datatype/Units	Allowed Range	Missing-Value Handling	Notes
ALLSKY_SFC_SW_DWN	All-sky global horizontal irradiance (target)	Continuous, kWh/m ² /day	0–9	–999 sentinel → NA; ≤ 3 -day interpolation (none required)	Dependent variable (Y)

Variable	Definition	Datatype/Units	Allowed Range	Missing-Value Handling	Notes
T2M	Air temperature at 2 m	Continuous, °C	5–45	Same as above; 0 missing	Predictor
RH2M	Relative humidity at 2 m	Continuous, %	0–100	Same as above; 0 missing	Predictor
CLOUD_AMT	Cloud amount	Continuous, %	0–100	Same as above; 0 missing	Predictor
PRECTOTCORR	Corrected total precipitation	Continuous, mm/day	0–400	Same as above; 0 missing	Predictor
WS2M	Wind speed at 2 m	Continuous, m/s	0–25	Same as above; 0 missing	Predictor
doy_sin, doy_cos	Cyclical day-of-year encoding	Continuous, [−1, 1]	—	Deterministic (no missingness)	Engineered predictor
season	Harmattan-dry / major-wet / minor-dry / minor-wet	Categorical	4 levels	Deterministic	Predictor / stratifier
ghi_lag1 ... ghi_lag7	Lagged GHI (t−1 ... t−7)	Continuous, kWh/m ² /day	0–9	First 7 days per site undefined; rows dropped in modeling	Engineered predictor
ghi_roll7, ghi_roll30	Rolling-mean GHI (7-, 30-day)	Continuous, kWh/m ² /day	0–9	First 7/30 days undefined; rows dropped in modeling	Engineered predictor
site, zone	Location / climatic-zone identifier	Categorical	5 / 5 levels	Deterministic	Grouping variable

GitHub Data Availability Statement

The raw and processed datasets are available in the GitHub repository (Adjei-Mensah, 2026) under:

- Raw data: data/raw/ (one NASA POWER CSV per site, plus download_log.txt with row counts and SHA-256 digests).
- Processed/clean data: data/processed/ (per-site feature files and the combined all_sites_daily.csv).
- Code/notebooks: src/ (config, download, validation, cleaning modules) and notebooks/ (01–07); metrics tables and figures in results/.

Screenshots / Evidence

Figure 1 embeds the dataset preview required by the template: the first rows of the Accra raw file exactly as retrieved from NASA POWER (all five site files share the same layout). The

full row-count and calendar-continuity report, missing-value tables, and physical-range checks are produced and displayed in notebook 02 (02_data_validation.ipynb), and the repository tree structure appears in Appendix A. Table 3 summarizes the validation outcome that this evidence supports. The acceptance rule for missingness (below 2% per variable) is intentionally stricter than the roughly 5% completeness bars common in climate-data quality control; in the event the archive is complete, so the rule is not binding.

Figure 1

Dataset Preview: Raw NASA POWER Daily Record for Accra

Raw NASA POWER daily record, Accra (first 10 of 7,305 rows)

YEAR	MO	DY	ALLSKY_SFC_SW_DWN	T2M	RH2M	CLOUD_AMT	PRECTOTCORR	WS2M
2005.0	1.0	1.0	5.26	26.82	85.64	47.32	5.05	1.76
2005.0	1.0	2.0	5.87	26.32	71.6	12.33	1.08	2.12
2005.0	1.0	3.0	5.7	26.75	77.16	1.61	0.01	1.35
2005.0	1.0	4.0	5.87	27.05	77.45	5.2	0.26	1.37
2005.0	1.0	5.0	4.74	27.02	83.28	51.57	1.25	1.47
2005.0	1.0	6.0	5.0	27.0	76.57	41.14	0.58	1.66
2005.0	1.0	7.0	4.66	25.46	69.32	22.14	0.0	2.21
2005.0	1.0	8.0	3.86	25.03	60.78	27.2	0.0	3.97
2005.0	1.0	9.0	4.86	24.46	58.97	7.53	0.0	3.91
2005.0	1.0	10.0	5.18	25.15	59.15	25.34	0.0	3.24

Note. First 10 of 7,305 rows. Generated in notebook 02; results/figures/fig_02_1_dataset_preview.png.

Table 3

Data Validation Summary (All Five Sites)

Check	Expectation	Result	Verdict
Row count per site	7,305 (2005-01-01 → 2024-12-31)	7,305 at all five sites	PASS
Calendar continuity	No missing or duplicate dates	0 missing, 0 duplicates	PASS
Missing values (–999 sentinels)	< 2% per variable	0 missing in all 6 variables × 5 sites	PASS
Physical ranges	All values in plausible envelopes	All in range (site GHI 0.51–7.71 kWh/m ² /day)	PASS

Analysis

Data Cleaning

Cleaning (src/clean_data.py, verified in notebook 02) proceeds in four steps: (a) NASA POWER's -999 missing-value sentinels are converted to NA; (b) each site's series is re-indexed onto a continuous daily calendar; (c) gaps of up to 3 days would be linearly interpolated, in the event, the archive contained zero missing values, so no imputation was applied anywhere; and (d) features are engineered (GHI lags 1–7, 7- and 30-day rolling means, sine/cosine day-of-year encoding, and a four-level season label reflecting Ghana's bimodal southern regime, with December–February labeled Harmattan-dry). Values outside physical envelopes were screened (Table 3), and none were found; consequently, no outlier treatment beyond range screening was required, and duplicates were absent. Table 4 records the cleaning log in the required format. The 3-day interpolation cap is deliberate: filling longer gaps would fabricate exactly the multi-day low-sun spells whose lengths drive the RQ4 sizing analysis, so longer gaps would instead be left missing and excluded from affected computations.

Table 4

Data Cleaning Log

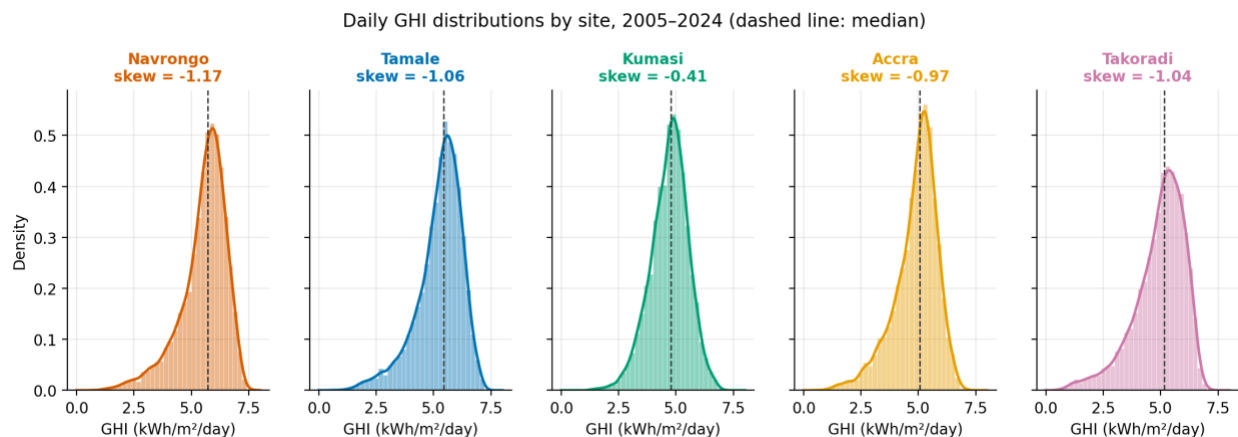
Issue	Variables Affected	Detection Method	Treatment Applied	Rationale
Missing values	All 6 parameters	–999 sentinel scan + NA count per site	None required (0 missing; ≤3-day interpolation available)	NASA POWER delivered complete series 2005–2024
Outliers	All 6 parameters	Physical-range envelopes (e.g., GHI 0–9 kWh/m ² /day)	None required (0 out of range)	Screen for sensor/reanalysis artifacts without distorting real extremes
Duplicates	Date index	Duplicate-date scan per site	None required (0 found)	Guarantee one observation per site-day
Undefined lag/rolling rows	Engineered features	NA scan after feature engineering	Rows dropped at modeling time only	Preserve full series for EDA; avoid fabricating history

EDA Results

All figures below are generated in notebook 03 and saved under results/figures/; each is interpreted in the text as required. Summary statistics ground the analysis: mean daily GHI declines from north to south: Navrongo 5.53, Tamale 5.25, Takoradi 4.96, Accra 4.89, and Kumasi 4.71 kWh/m²/day, with the moist-forest site, not the coast, the least sunny on average. Takoradi is the most variable site (SD = 1.09 kWh/m²/day; 5th percentile 2.79, the lowest of the five), a first indication that coastal cloudiness rather than average resource is the binding constraint for both forecasting and sizing. The subsections move from univariate distributions to bivariate relationships and then to temporal and extreme-value structure.

Figure 2

Daily GHI Distributions by Site, 2005–2024



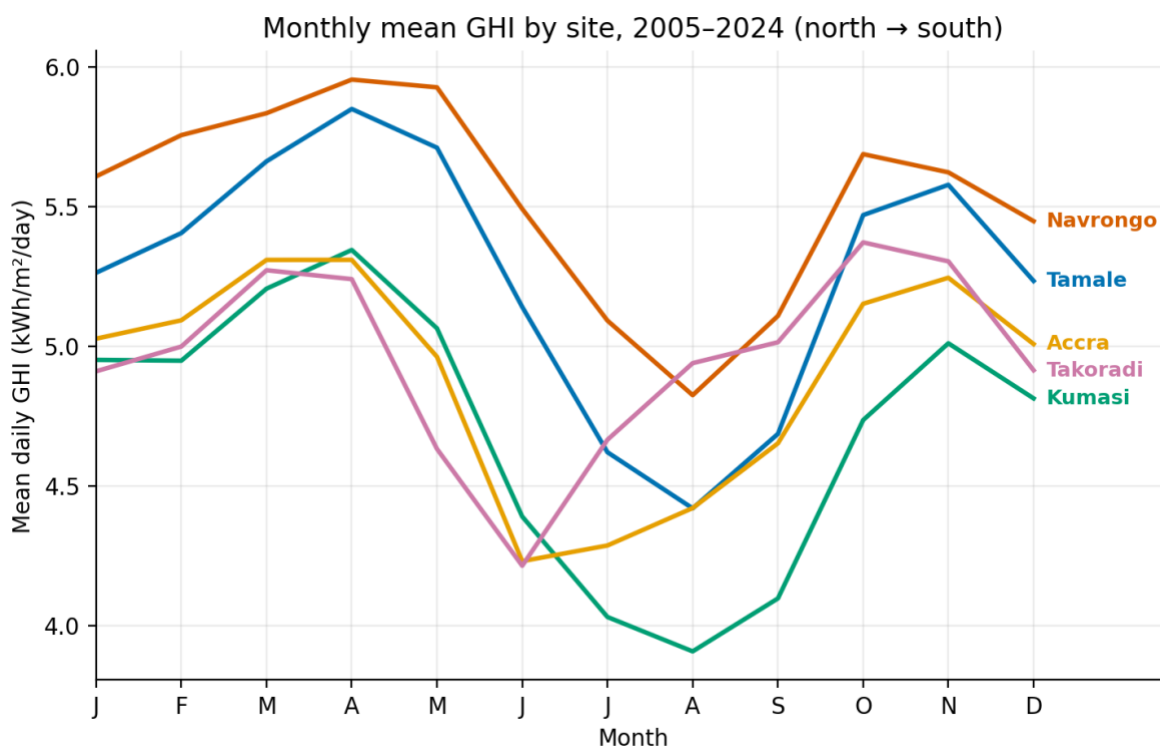
Note. Histograms with kernel density estimates; dashed line marks the median. Notebook 03, results/figures/fig_03_6_ghi_histograms.png; skewness values shown per panel.

Figure 2 shows the univariate distribution of the target. Every site is left-skewed: the typical day sits near 5–6 kWh/m²/day while the risk concentrates in a long lower tail of cloudy days, with skewness strengthening from Kumasi (−0.41, the most symmetric and narrowest distribution) to Navrongo (−1.17). Two implications follow. For forecasting (RQ1), squared-error metrics are dominated by tail days, so a model that tracks only the seasonal mean can look

adequate on average while missing the days that matter most. For sizing (RQ4), the mean overstates the dependable resource; a reliability-constrained design must be driven by the lower tail, which is precisely what the planned LOLP simulation does.

Figure 3

Monthly Mean GHI by Site, 2005–2024



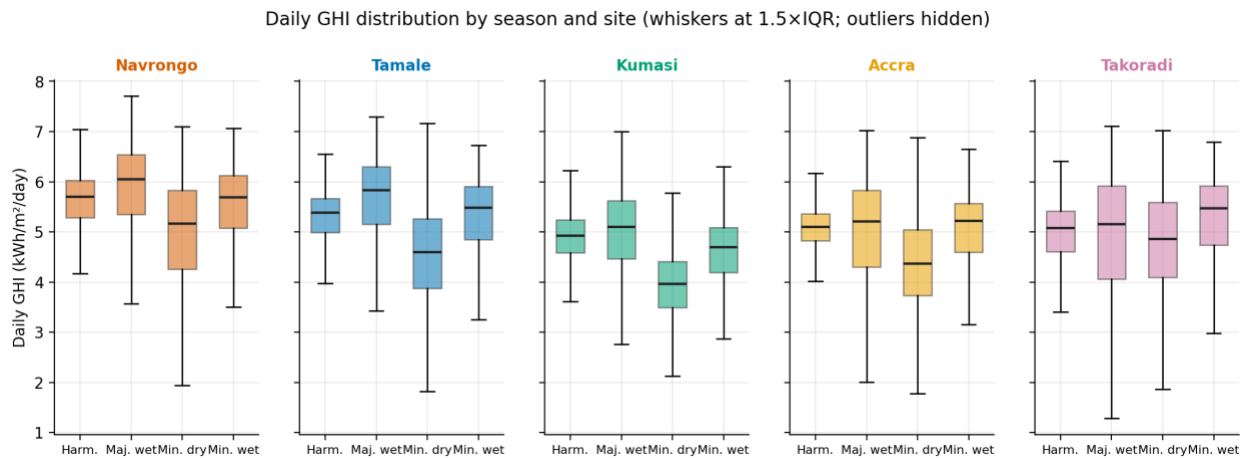
Note. Sites ordered north to south. Data: NASA POWER (2005–2024); notebook 03, results/figures/fig_03_1_monthly_climatology.png.

As Figure 3 shows, all five sites share a bimodal seasonal cycle with a mid-year minimum, but its depth differs sharply by zone: the northern savannah sites peak in March–April (Navrongo ≈ 6.0 kWh/m²/day) and dip only to about 4.8 in the July–August monsoon, whereas the southern sites fall much further (Kumasi bottoms near 3.9 in August). The persistent 0.8–1.0 kWh/m²/day gap between Navrongo and Kumasi across most months is the north–south gradient

over which RQ3 stratifies. The Harmattan months (December–February) remain comparatively sunny at all sites: dust attenuation moderates, but does not remove, the dry-season advantage.

Figure 4

Daily GHI Distribution by Season and Site

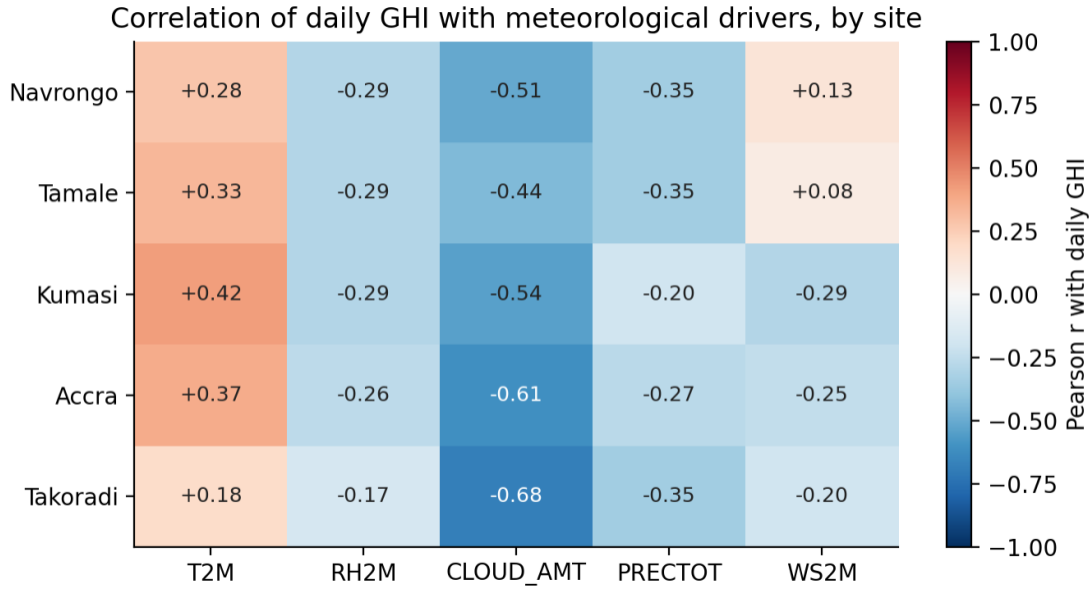


Note. Whiskers at $1.5 \times \text{IQR}$; outliers hidden for legibility. Notebook 03, fig_03_2_seasonal_boxplots.png.

Figure 4 shows that seasonal contrast grows from north to south. At Navrongo and Tamale, the four seasonal distributions overlap heavily; at Kumasi, Accra, and Takoradi, the wet-season boxes shift visibly downward and widen, with the coastal sites showing the longest lower tails. For RQ3, this suggests that forecast errors will be season-dependent mainly in the south; for RQ4, it shows that the wet-season floor, not the annual mean, is what a reliability-constrained system must be sized against.

Figure 5

Correlation of Daily GHI With Meteorological Drivers, by Site

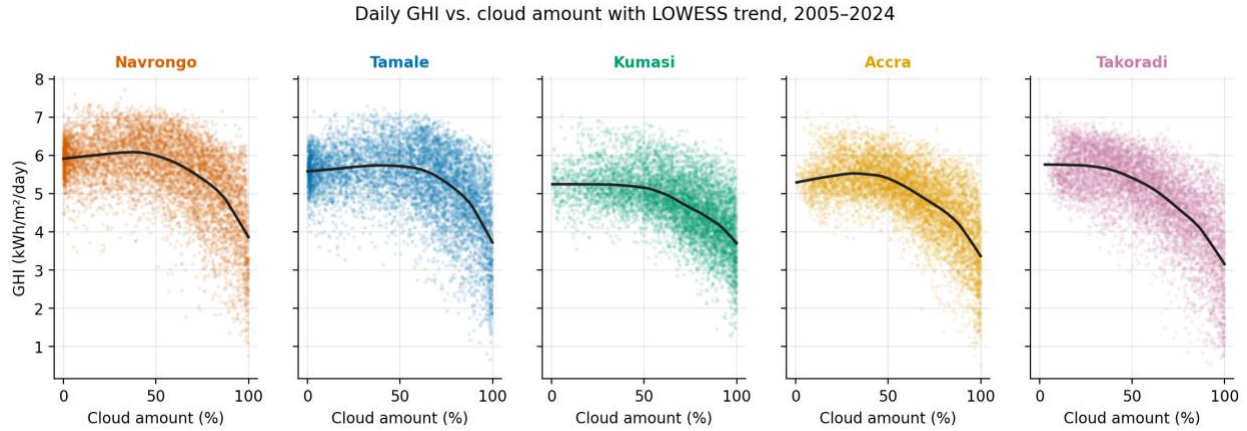


Note. Pearson r, full 2005–2024 record. Notebook 03, fig_03_3_driver_correlations.png; values also in results/eda_driver_correlations.csv.

Figure 5 addresses RQ2’s groundwork. Cloud amount is the dominant (negative) correlate of daily GHI at every site, strengthening toward the coast: $r = -.44$ at Tamale but $-.68$ at Takoradi. Precipitation and relative humidity show moderate negative associations ($r \approx -.17$ to $-.35$) and temperature moderate positive ones ($r = .18-.42$). Wind speed is the only sign-flipping driver: weakly positive in the north (consistent with dry Harmattan airflow) and negative in the south (consistent with onshore monsoon flow bringing cloud). These are associations, not importances; formal SHAP analysis follows in notebook 06.

Figure 6

Daily GHI Versus Cloud Amount With LOWESS Trend, 2005–2024

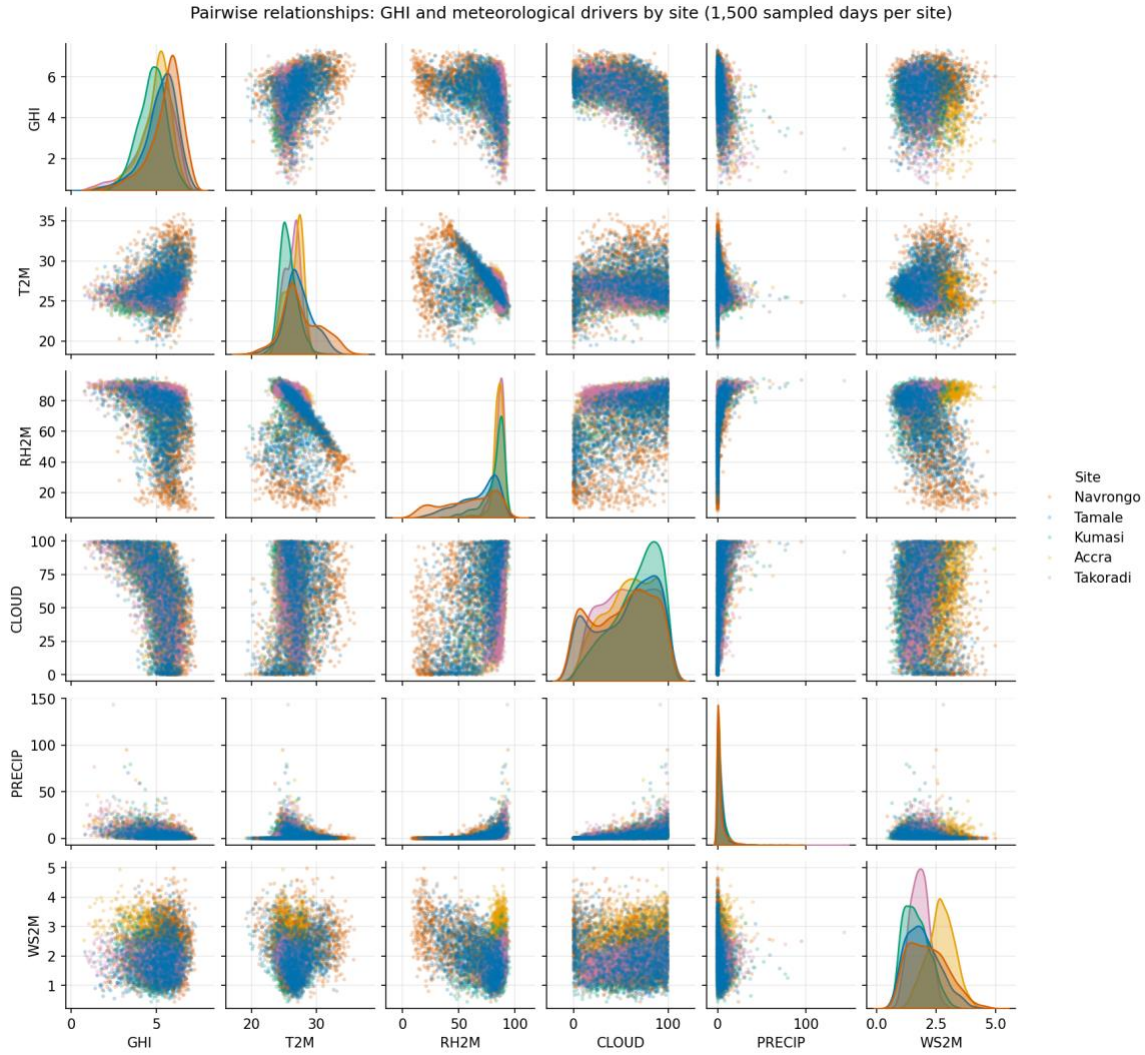


Note. Points rasterized at alpha 0.08; black curve is the locally weighted (LOWESS) trend per site. Notebook 03, results/figures/fig_03_7_ghi_vs_cloud.png.

Figure 6 shows the shape of the dominant bivariate relationship. The LOWESS trend is nearly flat up to roughly 40–50% cloud cover and then falls steeply, with the vertical spread widening as skies cloud over; the association is closer to a threshold effect than to the linear relationship the coefficients in Figure 5 summarize. This is direct empirical justification for preferring nonlinear tree-ensemble models over linear regression (RQ1), it explains why the coastal sites are the hardest to forecast (they spend the most days in the steep, high-variance region of the curve), and it sharpens the RQ2 expectation that cloud amount’s SHAP contributions will concentrate on high-cloud days.

Figure 7

Pairwise Relationships Among GHI and All Meteorological Drivers, by Site



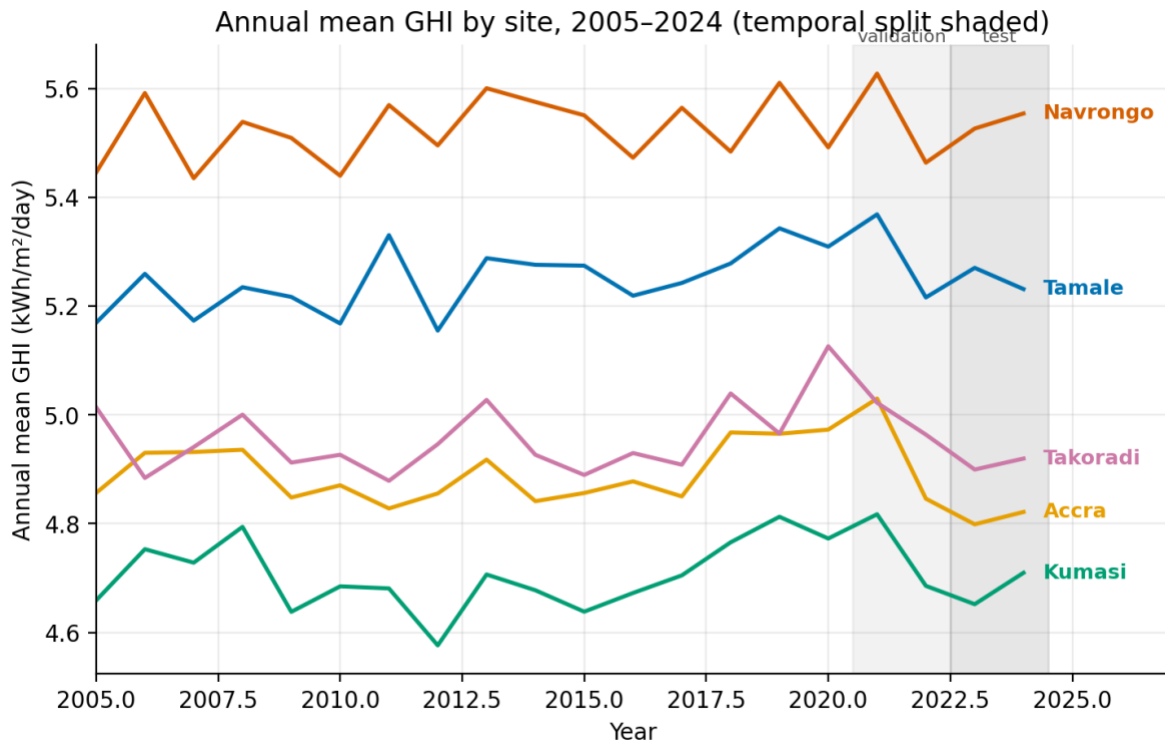
Note. Seaborn pair plot; 1,500 randomly sampled days per site (fixed seed), kernel density estimates on the diagonal. Notebook 03, results/figures/fig_03_9_pairplot.png. The sample size is for plot legibility only; all statistics elsewhere in this report use the full record.

Figure 7 assembles the univariate diagonal and every bivariate combination in one view and adds three observations the single-relationship figures cannot show. First, the predictors are themselves inter-correlated: temperature and relative humidity form a tight negative band, and cloud amount rises with both humidity and precipitation, so multicollinearity among drivers is real; the VIF screening planned for RQ2 is warranted, and importance rankings should come from permutation and SHAP methods rather than raw regression coefficients. Second, the sites separate most clearly in the driver panels rather than the GHI panels: wind speed nearly isolates

Accra (the windiest site, mean 2.7 m/s) from Kumasi (the calmest, 1.6 m/s), and the humidity diagonal splits the broad northern distributions from the coastal sites' narrow peaks near saturation; this is the regime signal available to site-aware models. Third, no bivariate panel shows structure stronger than GHI versus cloud amount, confirming the threshold pattern of Figure 6 as the dominant relationship the forecasting models must capture. Precipitation is heavily zero-inflated, which cautions against treating its linear correlation as the full story.

Figure 8

Annual Mean GHI by Site, 2005–2024, With the Temporal Split Shaded



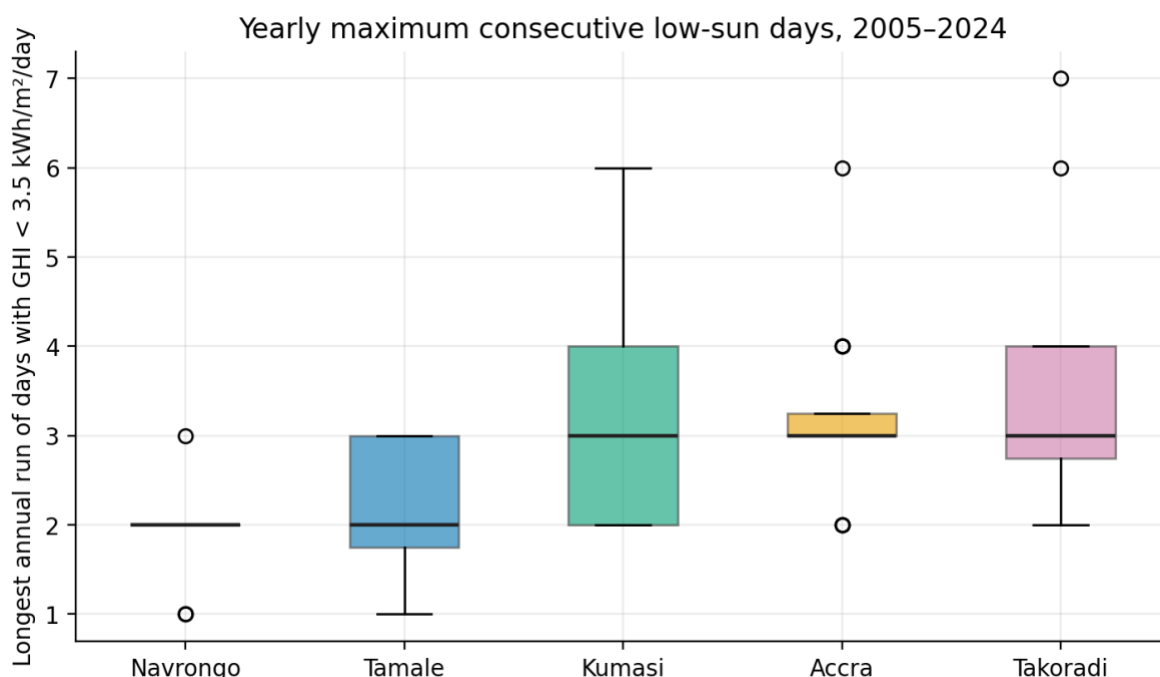
Note. Light shading: validation (2021–2022); darker shading: test (2023–2024). Notebook 03, fig_03_4_annual_means.png.

Figure 8 shows annual means fluctuating within a narrow ± 0.2 – 0.3 kWh/m²/day band with no visible monotonic trend and a stable site ranking. The validation and test windows sit within the historical range; neither anomalously sunny nor cloudy, so the fixed temporal split

provides a fair evaluation period, and the full 20-year record is defensible input for the RQ4 sizing simulation.

Figure 9

Yearly Maximum Consecutive Low-Sun Days ($GHI < 3.5 \text{ kWh/m}^2/\text{day}$), 2005–2024



Note. Notebook 03, fig_03_5_low_sun_runs.png; summary in results/eda_low_sun_runs.csv.

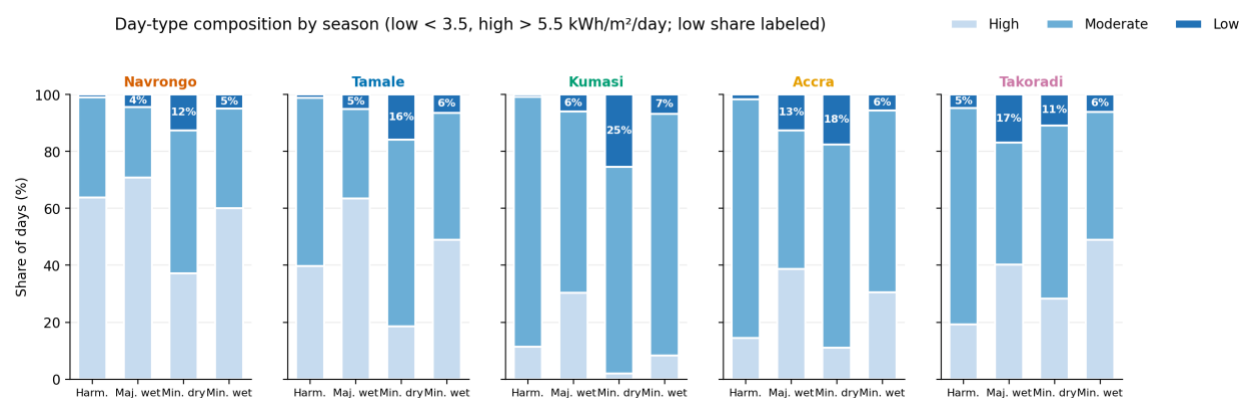
Figure 9 provides direct groundwork for RQ4. Low-sun spells are roughly twice as severe in the south: the median longest annual run below $3.5 \text{ kWh/m}^2/\text{day}$ is 2 consecutive days at Navrongo and Tamale versus 3 days at the three southern sites, and the 20-year worst case grows from 3 days in the north to 6–7 days at Kumasi, Accra, and Takoradi. Total exposure follows the same gradient (about 18 low-sun days per year at Navrongo versus about 37 at Takoradi). A fixed days-of-autonomy rule calibrated in the north would therefore under-protect a coastal household, exactly the zone-specific sizing gap this project addresses.

The $3.5 \text{ kWh/m}^2/\text{day}$ threshold is anchored in the reference design rather than chosen freely: a system sized on worst-month peak sun hours of 4.0 at a performance ratio of 0.75 fails

to cover its daily load from same-day generation once GHI falls below approximately 3.5, making this the level at which the battery, not the array, carries the household. The 5.5 kWh/m²/day bound used in the day-type analysis marks comfortable surplus operation. Both cutoffs are classification aids for description only; the sizing simulation itself uses the continuous record, so no result depends on the exact values.

Figure 10

Day-Type Composition by Site and Season (Low < 3.5, High > 5.5 kWh/m²/day)



Note. 100% stacked shares of low, moderate, and high GHI days; low-sun share labeled. Notebook 03, results/figures/fig_03_8_day_type_composition.png; table in results/eda_day_type_composition.csv.

Figure 10 converts the distributions into the proportions that matter for sizing and shows when the low-sun exposure occurs. Low-sun days are rare everywhere in the Harmattan (0–2% of days) but concentrate in the mid-year cloudy season with sharply different intensity by zone: the minor-dry (July–August) low-sun share reaches 25% at Kumasi and 18% at Accra, versus 12% at Navrongo, while Takoradi’s burden peaks in the major wet season (17%). Conversely, Navrongo enjoys high-sun conditions on 60–71% of days outside the minor dry season, whereas Kumasi rarely exceeds 5.5 kWh/m²/day in any season. Southern systems must therefore be engineered around a season in which roughly one day in four to six under-produces; a single national autonomy rule cannot serve both Navrongo and Kumasi (RQ4).

Table 5*EDA Insight Summary*

Figure/Table	What it Shows	Key Insight	Why it Matters (RQ link)	Decision/Next Step
Summary statistics	Per-site GHI distribution	N–S mean gradient (5.53 → 4.71); Takoradi most variable	Zone contrast motivating the design (RQ1/RQ3)	Keep all 5 sites; report per-site metrics
Figure 3	Monthly climatology	Bimodal cycle; wet-season trough far deeper in south	Seasonality is the dominant signal (RQ1, RQ3)	Retain calendar harmonics in all models
Figure 4	Seasonal distributions	Season effects strongest at coastal/forest sites	Predicts season-dependent errors (RQ3)	Stratify metrics by season in notebook 06
Figure 5	Driver correlations	Cloud amount dominates (–.44 to –.68); wind flips sign N vs. S	Prior expectation for importance (RQ2)	Compare against SHAP rankings in notebook 06
Figure 8	Annual means	No trend; validation/test years typical	Split fairness; 20-yr record valid (RQ4)	Proceed with fixed temporal split
Figure 9	Low-sun runs	Worst runs 3 d (north) vs. 6–7 d (south)	Zone-specific autonomy need (RQ4)	Sizing simulation must be per-zone
Figure 2	Univariate GHI distributions	All sites left-skewed (–0.41 to –1.17); risk in lower tail	Tail days dominate RMSE (RQ1); mean overstates resource (RQ4)	Evaluate tail-day errors; size on the lower tail
Figure 6	GHI vs. cloud amount (bivariate)	Nonlinear threshold shape: flat to ~40–50% cloud, then steep decline	Justifies nonlinear tree ensembles (RQ1)	Retain RF/XGBoost over linear models
Figure 10	Day-type composition by season	Minor-dry low-sun share 25% Kumasi, 18% Accra vs 12% Navrongo	Low-sun exposure is seasonal and zonal (RQ3/RQ4)	Size against the worst season, per zone
Figure 7	Pair plot, all variable pairs	Drivers inter-correlated (T2M–RH2M band); sites separate in driver panels	Multicollinearity: use permutation/SHAP, VIF screening (RQ2)	Rank importance with SHAP, not coefficients

Minimum Sample Size Computation per Research Question

The dataset is a fixed archive rather than a survey to be fielded, so this subsection verifies that the available data volume is adequate for each research question, using confidence-interval (CI) precision and power-of-test methods as appropriate. Throughout, the significance level is $\alpha = .05$ ($z = 1.96$) and, where power is used, power is 90% ($z\beta = 1.28$), deliberately stricter than the conventional 80%; $\rho = .10$ and $d = 0.10$ correspond to Cohen’s (1988) thresholds for small effects, so even weak meteorological drivers and modest accuracy gains remain detectable. The margin of error $E = 0.05$ kWh/m²/day is approximately 1% of Ghana’s mean daily GHI, and $\sigma = 1.0$ kWh/m²/day approximates the day-to-day GHI variability observed at Ghanaian stations

(Quansah et al., 2022). Because daily observations are serially dependent, these independent-observation formulas serve as approximate benchmarks for precision and power; final uncertainty estimates and hypothesis tests use time-series-aware procedures (moving-block bootstrap; corrected Diebold–Mariano tests). Table 6 summarizes the calculations.

RQ1a (CI precision): To estimate mean forecast error with a margin of error $E = 0.05$ kWh/m²/day, taking $\sigma = 1.0$ kWh/m²/day as a deliberately conservative bound on error variability, the required number of daily observations per site is

$$n = \left(\frac{z \times \sigma}{E}\right)^2 = \left(\frac{1.96 \times 1.0}{0.05}\right)^2 \approx 1,537$$

RQ1b (power of test, paired comparison): Models are compared on the same test days, a paired design on daily error differences. To detect a small standardized effect of $d = 0.10$, the required number of paired test days is

$$n = \left(\frac{z_{\alpha/2} + z_{\beta}}{d}\right)^2 \approx 1,051$$

RQ2 (power of test, correlation): To detect a weak correlation of $\rho = .10$ at $\alpha = .05$ with 90% power via the Fisher z transformation, where $C(\rho) = 0.1003$, the required number of observations is

$$n = \left(\frac{z_{\alpha/2} + z_{\beta}}{C(\rho)}\right)^2 + 3 \approx 1,047$$

RQ3 (power of test, two-sample): To compare wet-season and dry-season forecast errors with effect size $d = 0.20$, the required number of days per season group is

$$n = 2\left(\frac{z_{\alpha/2} + z_{\beta}}{d}\right)^2 \approx 526$$

RQ4 (CI precision, proportion): The LOLP is a proportion, the share of observed historical days on which the modeled system would fail to serve the load. To estimate an LOLP

near $p = .01$ within $E = .005$ at 95% confidence, the required number of historical days evaluated per zone–configuration is

$$n = \frac{z^2 p(1 - p)}{E^2} \approx 1,522$$

Because loss-of-load days occur in runs rather than independently, interval estimates for the LOLP will additionally use year-block and moving-block bootstrap methods.

Table 6

Minimum Sample Size per Research Question

Research Question	Method Used	Key Parameters	Minimum Sample Size (N)
RQ1a	Confidence interval (mean)	$z = 1.96$, $\sigma = 1.0$ kWh/m ² /day, $E = 0.05$	1,537 days/site
RQ1b	Power of test (paired difference)	$\alpha = .05$, power = .90, $d = 0.10$	1,051 test days
RQ2	Power of test (correlation, Fisher z)	$\alpha = .05$, power = .90, $\rho = .10$	1,047 days
RQ3	Power of test (two-sample)	$\alpha = .05$, power = .90, $d = 0.20$	526 days/season group
RQ4	Confidence interval (proportion)	$z = 1.96$, $p = .01$, $E = .005$	1,522 days/zone

The binding requirement is $N = \max(\text{RQ1–RQ4}) = 1,537$ daily observations per site (about 4.2 years of daily data); the retrieved 20-year record provides 7,305 observations per site and 36,525 in total, exceeding that requirement by more than four times. The held-out test period provides about 730 paired days per site, below the 1,051-day benchmark for a single site; the requirement is met by pooling test days across the five sites (about 3,650 paired observations), with per-site comparisons reported descriptively, and the RQ3 per-season requirement is likewise met by pooling (about 1,825 test days per season group against the 526 required). Pooling is used only to increase inferential precision: because weather is spatially correlated across Ghana,

pooled tests resample time blocks jointly across sites so that cross-site dependence is preserved rather than assumed away, and site-level estimates remain the primary substantive results.

Modelling

Choice of Models With Justification

Model 1: Persistence (classical benchmark). The forecast equals the last observed value. Included because every forecasting claim must clear the weakest sensible reference (Voyant et al., 2017), and because published tropical effect sizes are typically quoted against persistence-type baselines (Qing & Niu, 2018).

Model 2: Day-of-year climatology (classical benchmark). The forecast equals the 2005–2020 training-window mean GHI for that calendar day. It captures the seasonal cycle, the dominant signal identified in EDA, and is the reference for the study’s $\geq 15\%$ RMSE-reduction success criterion (H1).

Model 3: Dynamic harmonic regression (DHR; SARIMA-class benchmark). A regression of GHI on $K = 3$ annual Fourier harmonics with AR(1) errors, estimated per site on training data (statsmodels SARIMAX). DHR handles the ≈ 365 -day cycle more efficiently than a large seasonal SARIMA structure (Hyndman & Athanasopoulos, 2021) and adds short-memory anomaly persistence on top of climatology. $K = 3$ is the smallest harmonic order that reproduces the bimodal annual cycle with asymmetric peaks (a single harmonic cannot produce two maxima); sensitivity to K remains checkable by AIC during final tuning.

Model 4: Random forest (ML). Tree ensemble robust to nonlinear interactions among engineered tabular features without distributional assumptions (Breiman, 2001); provides variable-importance machinery feeding RQ2.

Model 5: XGBoost (ML). Regularized gradient boosting, the standard for structured tabular prediction (Chen & Guestrin, 2016); its gain-based importances complement RQ2.

Model 6: LSTM (ML; pending). Recurrent network on 30-day input windows, able to carry information across multi-day weather episodes (Hochreiter & Schmidhuber, 1997); the main deep-learning precedent in a tropical setting is Qing and Niu (2018). The 30-day input window spans a full month of history, covering multi-day cloudy episodes and Harmattan spells, and mirrors the 30-day rolling-mean feature used by the tree models.

Features Included and Feature Engineering

For an h -day-ahead forecast issued at origin day t , every feature is measured on or before t (direct strategy; no look-ahead): today's GHI and its lags, rolling means through t , today's meteorological values, and deterministic calendar features. Contemporaneous target-day weather is never used as a forecast input. Table 7 lists the feature set used in the preliminary ML run. Lag depth and window lengths match the phenomena they summarize: one week of daily lags equals the longest forecast horizon, while the 7- and 30-day rolling means capture synoptic and monthly regimes respectively.

Table 7

Feature Set and Usage

Feature	Original/Engineered	Type	Reason for Inclusion	Used in Model(s)
GHI at origin t + lags 1–6	Engineered (lags)	Continuous	Short-memory anomaly persistence	RF, XGBoost (LSTM pending)
ghi_roll7, ghi_roll30	Engineered (rolling means)	Continuous	Smoothed recent regime (top importance, Fig. 12)	RF, XGBoost
T2M, RH2M, CLOUD_AMT, PRECTOTCORR, WS2M at t	Original	Continuous	Meteorological state at origin (EDA: cloud dominant)	RF, XGBoost
doy_sin, doy_cos	Engineered (cyclical encoding)	Continuous	Seasonal position, known for any future date	All models (Fourier terms in DHR)

Feature	Original/Engineered	Type	Reason for Inclusion	Used in Model(s)
Season label	Engineered (categorical)	Categorical	Stratification for RQ3 (not a forecast input)	Evaluation only

Evaluation Metrics

Point-forecast accuracy uses the root-mean-square error (RMSE), the mean absolute error (MAE), the coefficient of determination (R^2), and forecast skill relative to a baseline; all are computed on the validation window (2021–2022) in this report and will be recomputed with confidence intervals on the untouched test window (2023–2024) for the final report.

$$RMSE = \sqrt{\frac{1}{n} \sum_t (y_t - \hat{y}_t)^2}$$

$$MAE = \frac{1}{n} \sum_t |y_t - \hat{y}_t|$$

$$R^2 = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \bar{y})^2}$$

$$Skill = 1 - \frac{RMSE_{model}}{RMSE_{baseline}}$$

Worked calculation. At Accra (1-day horizon, validation window), the day-of-year climatology baseline achieves $RMSE \approx 0.883$ kWh/m²/day, and the tuned random forest achieves $RMSE \approx 0.820$, giving $Skill = 1 - 0.820/0.883 = 0.071$ (7.1%). Against the study’s H1 success criterion of $Skill \geq 0.15$, this preliminary result falls short (see Preliminary Results for interpretation). System-reliability metrics for RQ4 (the LOLP, together with the loss-of-power-supply probability and the expected energy not served) will be defined and reported with the sizing simulation in the final report.

Preliminary Results

Results below use the strict temporal split from the study design: train 2005–2020, validation 2021–2022, with the 2023–2024 test window untouched until the final statistical comparison (notebook 06). Metrics tables are reproducible from results/baseline_metrics.csv and results/ml_vs_dhr_1day.csv in the repository. The split is approximately 80/10/10 in chronological order, and each held-out window contains two full annual cycles so that every season appears at least twice in both validation and test.

RQ1: Forecasting Accuracy: Classical Baselines

Table 8 reports validation RMSE for the three classical baselines at both horizons (full RMSE/MAE/R² detail in the repository). Three findings stand out. First, persistence is the weakest baseline everywhere, with negative R² at both horizons; daily GHI is too noisy for “tomorrow equals today.” Second, DHR is the best classical model at every site and both horizons (1-day RMSE 0.699–0.992 kWh/m²/day), edging climatology by roughly 2–5% because its AR(1) term ($\phi = 0.14$ –0.23) captures short-lived anomalies; by 7 days $\phi^7 \approx 0$ and DHR collapses onto climatology, as the near-identical 7-day scores show. Third, forecastability tracks climate: Kumasi is the easiest site (RMSE 0.699) and Takoradi the hardest (0.992), consistent with the EDA finding that coastal cloudiness drives variability.

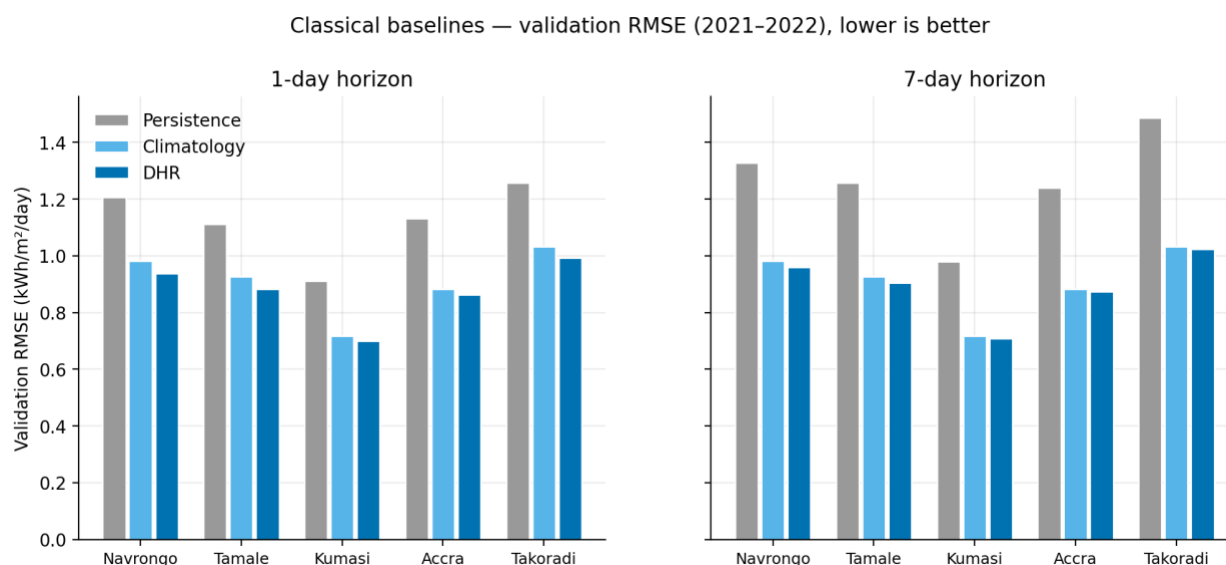
Table 8

Classical Baselines — Validation RMSE (kWh/m²/day), 2021–2022

Site	Persistence (1-d)	Climatology (1-d)	DHR (1-d)	Persistence (7-d)	Climatology (7-d)	DHR (7-d)
Navrongo	1.207	0.981	0.939	1.329	0.981	0.959
Tamale	1.111	0.928	0.882	1.257	0.928	0.905
Kumasi	0.911	0.717	0.699	0.980	0.717	0.709
Accra	1.133	0.883	0.863	1.240	0.883	0.874
Takoradi	1.257	1.033	0.992	1.487	1.033	1.023

Figure 11

Classical Baselines — Validation RMSE by Site, Model, and Horizon



Note. Notebook 04, fig_04_1_baseline_rmse.png.

RQ1: Preliminary ML Versus Classical (1-Day Horizon)

Random forest and XGBoost were fitted per site with small grids under expanding-window time-series cross-validation on the training window only (notebook 05). Table 9 and Figure 12 compare them with DHR, the strongest classical reference. Random forest edges XGBoost at all five sites and improves on DHR at every site, but the margins are modest: 0.5% (Navrongo) to 5.0% (Accra) RMSE reduction, and against the H1 reference baseline (day-of-year climatology), preliminary skill is 4.3–8.5%, short of the $\geq 15\%$ success criterion.

Table 9

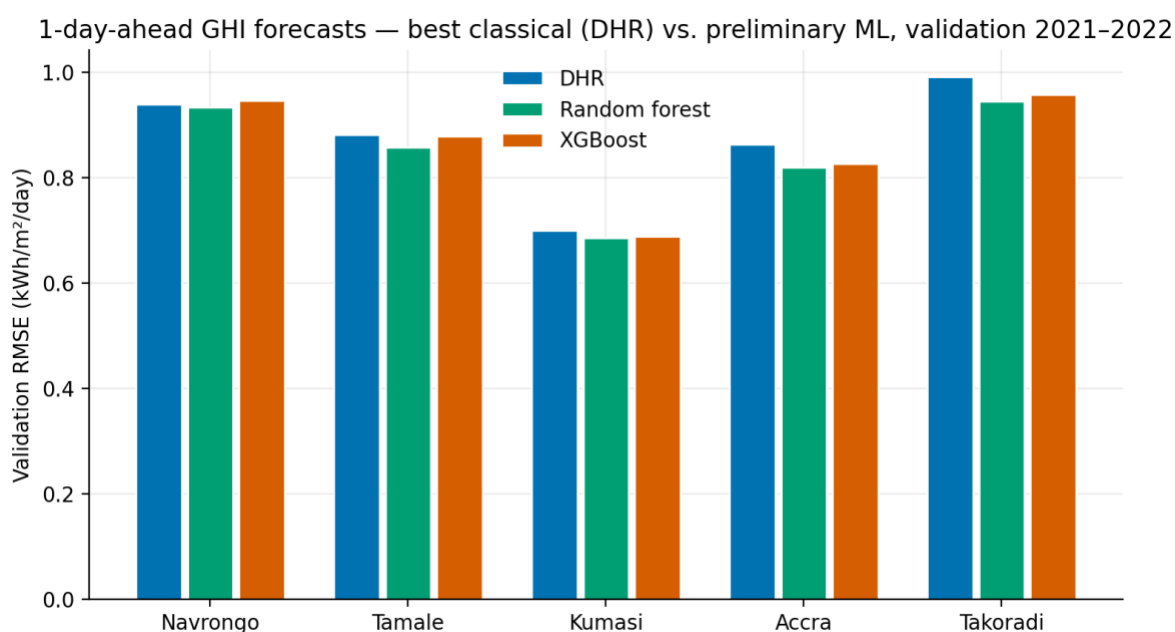
Preliminary Model Performance Comparison (1-Day Horizon, Validation 2021–2022)

Site	Model	Feature Set	Validation Approach	RMSE	MAE	Skill vs. DHR	Key Takeaway
Navrongo	Random forest	16 tabular features	Temporal split + TS-CV tuning	0.934	0.670	+0.5%	Marginal gain in stable north

Site	Model	Feature Set	Validation Approach	RMSE	MAE	Skill vs. DHR	Key Takeaway
Navrongo	XGBoost	16 tabular features	Temporal split + TS-CV tuning	0.946	0.676	−0.7%	Does not beat DHR here
Tamale	Random forest	"	"	0.857	0.621	+2.8%	Modest gain
Kumasi	Random forest	"	"	0.686	0.501	+1.9%	Easiest site overall
Accra	Random forest	"	"	0.820	0.601	+5.0%	Largest ML gain
Takoradi	Random forest	"	"	0.945	0.718	+4.7%	ML helps most where cloud varies

Figure 12

1-Day-Ahead Validation RMSE — Best Classical (DHR) Versus Preliminary ML



Note. Notebook 05, fig_05_1_ml_vs_dhr.png; XGBoost bars shown alongside random forest.

Interpretation. Against persistence, the baseline used in much of the tropical literature, the random forest reduces RMSE by 23–28% (e.g., Accra 1.133 → 0.820), squarely in the range reported by Qing and Niu (2018). The gap between “skill versus persistence” and “skill versus DHR/climatology” is therefore expected rather than anomalous: most 1-day predictability in daily GHI is seasonal, and the strong classical baselines already capture it. The ML challenge in this setting is forecasting the anomaly, where gains are small but real and are largest at the

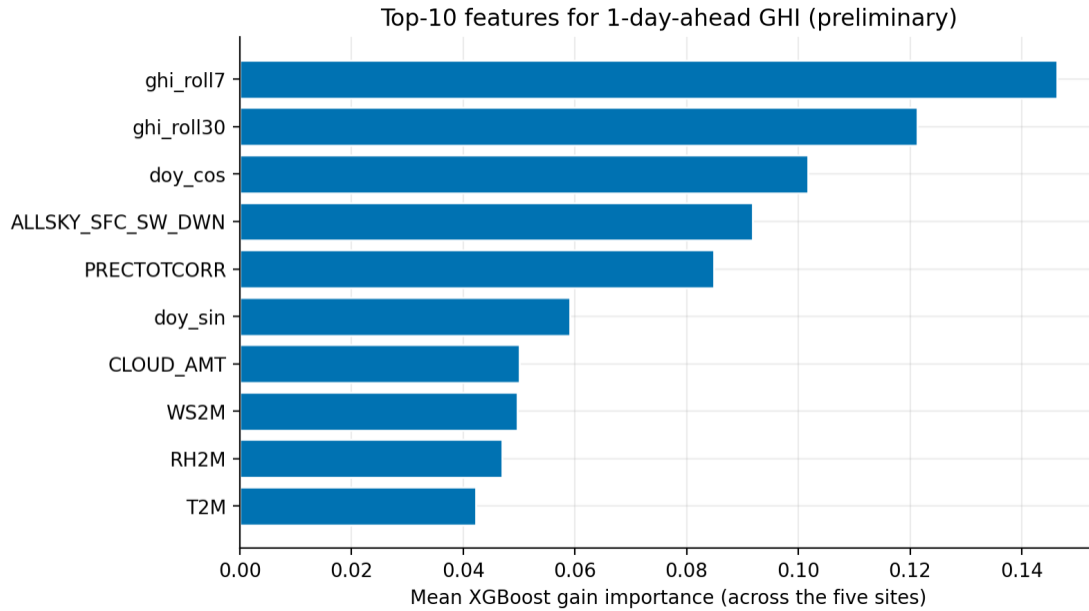
cloudier southern sites (Accra +5.0%, Takoradi +4.7% versus DHR), consistent with Lauret et al.'s (2015) finding that ML pays off in unstable-sky regimes. Reaching the 15% criterion will depend on the pending LSTM, richer tuning, and the 7-day horizon, and H1 will be formally tested on the untouched test window; the interim evidence keeps H1 open but flags it as at risk, an honest and defensible interim finding.

RQ2: Meteorological Drivers (Preliminary)

Two convergent strands of preliminary evidence address RQ2. Correlation analysis (Figure 5) identifies cloud amount as the dominant negative driver at all sites ($r = -.44$ to $-.68$), followed by precipitation and relative humidity, with wind speed sign-flipping between north and south. XGBoost gain-based importances (Figure 13) add the predictive view: recent GHI history (7- and 30-day rolling means, today's GHI) and the day-of-year harmonics carry the most weight, with precipitation and cloud amount the leading same-day meteorological inputs. Formal testing (Pearson/Spearman with Holm correction) and SHAP analysis on the test window remain scheduled for notebook 06; H2 (cloud amount and humidity strongest) is partially consistent with the correlation evidence so far, though precipitation currently rivals humidity.

Figure 13

Top-10 Features for 1-Day-Ahead GHI — Mean XGBoost Gain Importance Across Sites



Note. Notebook 05, fig_05_2_feature_importance.png; per-site values in results/ml_feature_importance_1day.csv.

RQ3: Seasonal and Zonal Differences (Groundwork)

Error stratification by season awaits the final models, but the EDA establishes the structure the stratification will test: seasonal contrast in the irradiance distribution itself is far stronger in the southern zones (Figure 4), and the classical-baseline results already show a stable zonal ordering of difficulty (Kumasi easiest, Takoradi hardest at every horizon). H3: that forecast RMSE differs significantly between wet and dry seasons with the largest degradation in the southern zones, therefore remains plausible and testable as designed.

RQ4: Off-Grid Sizing (Groundwork)

The sizing simulation has not yet run, but its key empirical input is now quantified: the distribution of consecutive low-sun days per zone (Figure 9). The 20-year worst runs, 3 days in the north versus 6–7 days in the south, already indicate that autonomy requirements differ materially across zones, in the direction hypothesized by H4 ($\geq 25\%$ zonal difference in required battery capacity). The deterministic starting grid for the LOLP search follows from a worst-month calculation: a 200 kWh/month household draws 6.67 kWh/day, and with worst-month

peak sun hours of 4.0 and a performance ratio of 0.75 the required array is approximately 2.4 kWp, with approximately 16.7 kWh of usable battery for two days of autonomy at 80% depth of discharge.

These constants are anchored rather than assumed freely. The performance ratio of 0.75 and the 80% usable depth of discharge sit within the ranges reviewed for standalone PV systems (Khatib et al., 2013), with inverter and wiring losses folded into the former; robustness to the performance-ratio assumption will be checked by repeating the backtest at 0.70 and 0.80. The worst-month value of 4.0 peak sun hours is taken from the observed record: the least-sunny monthly mean across the five sites is approximately 3.9 kWh/m²/day (Kumasi, August; Figure 3). The 2-day autonomy figure is the conventional installer rule of thumb, used here only as a starting grid that the simulation replaces with empirically observed low-sun runs.

Interim Limitations and Risks

- **H1 is at risk at the 1-day horizon.** Preliminary ML skill versus climatology (4.3–8.5%) is below the 15% criterion. Mitigation: LSTM and pooled multi-site training, broader tuning, 7-day horizon evaluation, and honest reporting of the outcome with Diebold–Mariano tests either way.
- **Satellite-derived data.** NASA POWER is a reanalysis product with known biases relative to ground pyranometers (Quansah et al., 2022); site-level uncertainty is acknowledged, and the strongest validation is in the north.
- **Zero-missing archive limits cleaning generality.** The interpolation machinery is untested on real gaps in this dataset; it remains in the pipeline for reproducibility on other sites.
- **Preliminary results are validation-window results.** All interim metrics use 2021–2022; conclusions may shift on the untouched 2023–2024 test window.
- **Data latency bounds operational use.** NASA POWER data arrive with a multiday latency, so the forecasts are positioned for planning and reserve scheduling rather than

real-time dispatch; operational adoption by GRIDCo or ECG would additionally require validating the forecasts against their own telemetry.

- **Compute and schedule risk.** LSTM training with repeated seeded runs is the most compute-intensive pending step; the design's stated prioritization (forecasting first, sizing second) protects the primary contribution if time becomes constrained. This prioritization is also the project's scope-management response to combining two substantial components in one capstone: the sizing study reuses the same processed record and pipeline rather than requiring new data or models.

Next Steps (for Final Report)

- Train the LSTM (30-day windows, early stopping, five seeded runs) and the pooled multi-site variant; extend RF/XGBoost to the 7-day horizon with broader hyperparameter search or tuning (notebook 05).
- Final evaluation on the untouched 2023–2024 test window: RMSE/MAE/R²/skill with moving-block-bootstrap confidence intervals; one-sided HLN-corrected Diebold–Mariano tests versus the classical baselines with Holm adjustment (RQ1; notebook 06). Covariance stationarity of the loss-differential series will be checked, the bootstrap block length will be set on the order of the error series' decorrelation time, and pooled tests will resample time blocks jointly across all five sites to preserve cross-site dependence; these diagnostics will be reported alongside the test results.
- SHAP and permutation-importance analysis with correlation testing under the Holm correction (RQ2; notebook 06).
- Seasonal (wet/dry/Harmattan) and zonal stratification of test-set errors with two-sample and nonparametric tests (RQ3; notebook 06).
- 20-year daily energy-balance sizing simulation per zone: grid search over array and battery capacity for $\text{LOLP} \leq 1\%$ and $\leq 5\%$ at the 100/200/300 kWh/month tiers; LPSP and EENS reported alongside; PV–battery trade-off frontiers (RQ4; notebook 07). Sizing results will be presented as one-page, zone-by-zone lookup tables usable without code or statistical training, with brief nontechnical documentation added to the repository.
- Synthesize zone-by-zone sizing tables and stakeholder recommendations; complete the final report and reproducibility packaging.

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Appendix A: Repository Structure and Reproduction

The repository referenced throughout this report (see the GitHub link following the title page) is organized as follows; the analysis in this report is reproduced by notebooks 02–05. This appendix is referenced from the Data Description and Analysis sections.

capstone-solar-ghana/	
├─ README.md	reproduction instructions
├─ requirements.txt	Python dependencies
├─ data/raw/	NASA POWER CSVs, one per site + download_log.txt
├─ data/processed/	cleaned, feature-engineered datasets
├─ notebooks/	01_download ... 07_offgrid_sizing
├─ src/	config.py, download_power.py, validate_data.py, clean_data.py
├─ results/	metrics tables (CSV), figures/, validation report
└─ reports/	interim & final report assets

Reproduction: `pip install -r requirements.txt`; `python src/download_power.py`; `python src/validate_data.py`; `python src/clean_data.py`; then run notebooks 02–05 in order (jupyter lab notebooks/).